

Class 25: Ratios and Rates

I. Ratios

Example: Don jumped 4 feet and Tim jumped 3 feet.

Three Ways to Write a Ratio

- i) $\frac{4}{3}$
- ii) 4:3
- iii) 4 to 3

① I have one dollar and you have six dollars. Write ratios of these two numbers in six different ways.

② Billy ate 20 sandwiches last month and Tom ate 11 sandwiches last month. Write ratios of these two numbers in six different ways.

③ The Kawasaki H2 supercharged motorcycle has 300 horsepower, while the ZX-6R motorcycle has 107 horsepower. Write ratios of these two numbers in six different ways.

Reducing Ratios to Lowest Terms

④ There were 8 gumballs in one bowl and 6 gumballs in another bowl. Write these two numbers as a ratio in lowest terms using a colon. Then, write the simplified ratio in two other ways.

⑤ A store sells a microphone for \$63, while another store sells the same model microphone for \$49. Write a ratio of these two numbers in lowest terms using a colon. Then, write this simplified ratio in two other ways.

Write the following ratios in lowest terms.

- ⑥ 8:10
- ⑦ 20 to 16
- ⑧ $\frac{6}{15}$
- ⑨ 45:25
- ⑩ 10 to 24
- ⑪ $\frac{28}{21}$
- ⑫ 18:9
- ⑬ 4 to 4
- ⑭ $\frac{8}{56}$
- ⑮ 26:26
- ⑯ 10 to 20
- ⑰ $\frac{36}{6}$

The Reverse Direction

⑮ The ratio of Mark's pocket change to Dillon's pocket change is $2:3$. If their combined pocket change is 30 cents, how much pocket change does Mark have?

⑯ The ratio of the areas of two different rectangles is 5 square meters to 7 square meters. If the total area of the two rectangles is 60 square meters, what are the areas of each rectangle?

⑰ Jake and Alice were playing a video game. The ratio of the points Jake scored to points Alice scored is $5:2$. If the total points that they scored is 63, how many points did Alice score?

⑱ If the ratio of the number of gallons of gasoline in Chelsea's car to the number of gallons of gasoline in Leo's car is 4 to 9, and the total number of gallons in both cars is 39, how many gallons of gasoline does Leo's car have?

The table below lists the costs of Amanda's school supplies. Answer the questions that follow the table.

Pencils	Paper	Binders	Dividers	Three-hole Punch
\$3	\$4	\$16	\$2	\$8

⑳ What is the ratio of the cost of dividers to the cost of paper?

㉑ Write the ratio of the cost of pencils to the cost of a three-hole punch.

㉒ Find the ratio of the cost of binders to the total cost of all Amanda's school supplies.

㉓ What is the ratio of the difference in cost between pencils and dividers to the difference in cost between binders and a three-hole punch?

㉔ Find the ratio of the sum of the costs of pencils and paper to the sum of the costs of binders and dividers.

More Examples of Ratios

②7 Jason scored 7 points, Linda scored 5 points, and Abbey scored 8 points. What is the ratio of the number of points that Abbey scored to the total number of points that all three people scored?

②8 Bobby's height increased from 60 inches to 65 inches. What is the ratio of his increase in height to his new height?

②9 Alan found three seashells, while Gordon found 14 seashells. What is the ratio of the total number of seashells that they both found to the number of seashells Alan found?

③0 The Rocket Burger contains three ounces of beef, whereas the Classic Burger contains four ounces of beef. What is the ratio of the volume difference between the beef in these two burgers and the volume of the beef in the Classic Burger?

③1 One watch costs \$40.23 and a second watch costs \$19.99. What is the ratio of the total cost of both watches to the cost of the first watch?

More Problems

③2 There are hammers and nails in a tool box. The ratio of the number of hammers to the total number of hammers and nails is $3:10$. What is the ratio of hammers to nails?

③3 The ratio of the number of foxes to the difference between the number of foxes and the number of coyotes is $4:3$. What is the ratio of the number of coyotes to the number of foxes?

③4 If the ratio of the total number of parakeets and finches to the number of parakeets is $8:5$, what is the ratio of the number of finches to the number of parakeets?

③5 The ratio of the difference between the number of puppies and the number of kittens to the number of kittens is $4:7$. What is the ratio of the number of kittens to the number of puppies?

Ratios Relating More than Two Numbers

③⑥ 6 red posters to 9 blue posters to 24 green posters.

③⑦ 10 arrows to 4 bows to 12 targets.

③⑧ 54 hotels to 81 homes to 45 apartments.

③⑨ If the ratio of Zane's number of books to Lisa's number of books to Ellic's number of books is $5:3:11$, and the sum of all their books is 57, how many books does each person have?

④⑩ The ratio of Edward's number of baseball cards to Steven's number of baseball cards to Casey's number of baseball cards is 7 to 10 to 4. If the combined number of all their cards is 105, how many cards does each person have?

II. Rates

Reduce the following rates to lowest terms.

④① 12 miles in 8 hours.

④② \$40 for 25 pics.

④③ 4 quarters for every 10 dimes.

④④ 18 cars for every 8 trucks.

④⑤ 81 bread sticks for every 54 cheese sticks.

④⑥ 21 miles on 14 gallons.

Unit Rates

④⑦ If there are 60 bunnies for every 6 kittens, how many bunnies are present for every 8 kittens?

④⑧ If there are 42 horses for every 7 donkeys, how many donkeys are there for every 18 horses?

④⑨ If a hiker travels 32 miles in 4 days, how many miles will he travel in 5 days?

⑤⑩ If it costs \$48 for every 6 ounces of a precious metal, how many ounces of precious metal could \$40 buy?

51) Nine toy airplanes cost \$63. How many planes could you buy with \$28?

52) If there are 100 chocolates in 5 boxes, how many chocolates are in 3 boxes?

53) If 4 pieces of candy cost \$3, how much would it cost to buy 7 pieces of candy?

54) If an armadillo crawls 5 miles in two hours, how many miles will he crawl in $3\frac{1}{2}$ hours?

55) There are 13 pounds of spaghetti for every 4 pounds of macaroni cheese. If there are 9 pounds of macaroni cheese, how many pounds of spaghetti are there?

56) If a recipe requires 4 ounces of flour for every 5 teaspoons of sugar, how many teaspoons of sugar are required for 22 ounces of flour?

57) If a construction company makes 8 feet of paved road for every 9 pounds of asphalt, how much asphalt would the company need to make 10 feet of road?

58) If a toy rocket can fly 2 miles on 3 ounces of fuel, how far could the rocket fly on 8 ounces of fuel?

59) If a deer travels 17 miles in 6 hours, how many miles will he travel in 4 hours?

60) If there are 22 ounces of cobalt for every 6 ounces of sulfur, how many ounces of cobalt would be present with one ounce of sulfur?

61) If 15 teaspoons of sugar are required for 7 pieces of cake, how many pieces of cake could be made with 16 teaspoons of sugar?

62) If a bird flies 11 kilometers in 16 minutes, how many kilometers would it fly in 7 minutes?